Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

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1. Executive Summary

The **Bluestock Mutual Fund Analytics Platform** is an end-to-end fintech analytics solution developed for Bluestock Fintech Pvt. Ltd. The project aims to consolidate mutual fund data from multiple public sources and transform it into actionable insights through data engineering, statistical analysis, and interactive visualization.

The platform processes **10 datasets covering 40 mutual fund schemes**, including NAV history, AUM, SIP inflows, investor transactions, portfolio holdings, scheme performance, and benchmark indices. Data is collected from sources such as **AMFI India, mfapi.in, NSE, and BSE**, then cleaned, validated, transformed, and stored using a structured SQLite database.

The analytical layer evaluates **fund performance, risk, and investor behaviour** using metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR. Exploratory analysis examines NAV trends, AUM growth, SIP inflows, investor demographics, geographic distribution, folio growth, and portfolio sector concentration.

The final solution is presented through a **four-page interactive BI dashboard** covering Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The dashboard provides KPIs, interactive filters, fund comparisons, investor insights, and market trend visualizations to support data-driven analysis.

Overall, the project demonstrates a complete **Extract** → **Transform** → **Load** → **Analyze** → **Visualize** workflow and provides a unified platform for understanding mutual fund performance, risk, industry trends, and investor behaviour. The project is intended for **educational and analytical purposes**, with investor transaction data being synthetically generated.

2. Data Sources & Datasets

All datasets used in this project are derived from publicly available, freely accessible sources. No proprietary or confidential data is used. The primary sources are:

Source	URL / API	Data Type	Update Freq
AMFI India	www.amfiindia.com	NAV, AUM, Folio, SIP	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
mfdata.in	mfdata.in/api/v1/schemes/	NAV + Expense Ratio	Daily
NSE India	nseindia.com/reports	Benchmark Index Prices	Daily
BSE India	bseindia.com	BSE SmallCap Index	Daily
AMFI Monthly Notes	amfiindia.com/research	Industry SIP & Flow Data	Monthly

Dataset Inventory

01_fund_master.csv ~40 rows

Master list of 40 real mutual fund schemes (AMFI codes, fund house, category, expense ratio,risk grade, fund manager)

02_nav_history.csv ~46,000 rows

Daily NAV for all 40 schemes from Jan 2022 to May 2026, anchored to real AMFI NAV values from mfapi.in

03_aum_by_fund_house.csv ~90 rows

Quarterly AUM (in Rs. crore) for 10 fund houses from 2022 to 2025, sourced from real AMFI quarterly reports

04_monthly_sip_inflows.csv 48 rows

Month-wise SIP inflow (Rs. crore), active SIP accounts, new registrations and SIP AUM - real AMFI Monthly Note data

05_category_inflows.csv ~144 rows

Net inflows by fund category (Large Cap, Mid Cap, Small Cap, ELSS, Liquid, etc.) for FY 2024-25

06_industry_folio_count.csv ~21 rows

Total mutual fund folios (in crore) broken by Equity, Debt, Hybrid -real AMFI published milestones

07_scheme_performance.csv ~40 rows

1yr/3yr/5yr returns, Sharpe, Sortino, Alpha, Beta, Max Drawdown, Std Dev -computed from NAV history

08_investor_transactions.csv ~32,000 rows

Simulated SIP + Lumpsum + Redemption transactions for 5,000 investors across Indian states with demographics

09_portfolio_holdings.csv ~320 rows

Top equity holdings (stock, weight %, sector) for equity mutual funds as of Dec 2025

10_benchmark_indices.csv ~8,000 rows

Daily closing values for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL Liquid & Gilt indices

3. ETL Design

The Bluestock Mutual Fund Analytics Platform follows an **Extract → Transform → Load (ETL)** architecture to consolidate fragmented mutual fund data into a structured analytical platform. The pipeline is designed to ingest data from multiple sources, clean and validate the datasets, store the processed information in a relational SQLite database, and make the resulting data available for analytics and dashboard development.

3.1 ETL Architecture

The overall workflow can be represented as:

Data Sources → Extraction → Data Cleaning & Validation → Transformation → SQLite Database → Analytics → Power BI Dashboard

The project addresses the problem of fragmented NAV, AUM, SIP, portfolio, benchmark, and investor data by consolidating these datasets into a single analytical system.

3.2 Extract

The extraction layer collects the project's **10 primary CSV datasets** covering fund information, NAV history, AUM, SIP inflows, category inflows, folios, scheme performance, investor transactions, portfolio holdings, and benchmark indices.

In addition, live NAV information is retrieved from the **mfapi.in REST API** for selected schemes. The project specifically requires fetching NAV data for HDFC Top 100 Direct and five selected schemes including SBI Bluechip, ICICI Prudential Bluechip, Nippon Large Cap, Axis Bluechip, and Kotak Bluechip.

The extracted files are initially stored in:

data/

└── raw/

This preserves the original source data before any transformations are applied.

3.3 Transform

The transformation stage performs data cleaning, validation, standardisation, and preparation for analysis.

Key transformations include:

- Parsing and standardising date columns.
- Sorting NAV records by amfi_code and date.
- Handling missing NAV observations caused by weekends and market holidays using forward-filling.
- Removing duplicate records.
- Validating that NAV and transaction amounts are greater than zero.
- Standardising transaction types into **SIP, Lumpsum, and Redemption**.
- Validating KYC status values.
- Converting scheme-performance fields to numeric data types.
- Checking expense ratios against the expected **0.1%–2.5%** range.
- Validating AMFI scheme codes between fund-master and NAV datasets.
- Preparing cleaned datasets for database loading.

The resulting datasets are stored in:

```
data/
└── processed/
```

3.4 Load

After transformation, the cleaned datasets are loaded into a **SQLite relational database** using Python, Pandas, and SQLAlchemy.

The database follows a star-schema approach with core dimension and fact tables such as:

dim_fund

dim_date

fact_nav

fact_transactions

fact_performance

fact_aum

The project specification defines amfi_code as the primary identifier for the fund dimension and uses foreign-key relationships from fact tables to connect financial observations to their corresponding funds and dates.

The database is stored under:

```
data/
└── db/
    └── bluestock_mf.db
```

3.5 Data Quality and Validation

Data quality checks are incorporated throughout the pipeline rather than being performed only after loading. The process validates data types, missing values, duplicate records, financial-value ranges, AMFI-code consistency, and transaction classifications.

For example, the NAV cleaning process validates that NAV values are positive and removes duplicate observations, while the AMFI validation step confirms that fund-master scheme codes exist in the NAV history.

3.6 ETL Outputs

The ETL process produces three main outputs:

1. **Cleaned CSV datasets** for analysis.
2. **SQLite database** for structured querying and downstream applications.
3. **Validated analytical data** used by the EDA, performance analytics, advanced analytics, and BI dashboard.

This architecture allows the same processed data to support Python-based analytics, SQL analysis, and Power BI visualization, creating a consistent data foundation throughout the project.

3.7 ETL Pipeline Tools

Stage	Tools
Extraction	Python, Pandas, Requests, mfapi.in
Transformation	Pandas, NumPy, Python
Validation	Pandas, NumPy, custom validation scripts
Database	SQLite
Database Interface	SQL Alchemy
Analytics	Python, Jupyter

Stage	Tools
Visualizations	Matplotlib, Seaborn, Plotly
Dashboard	Power BI

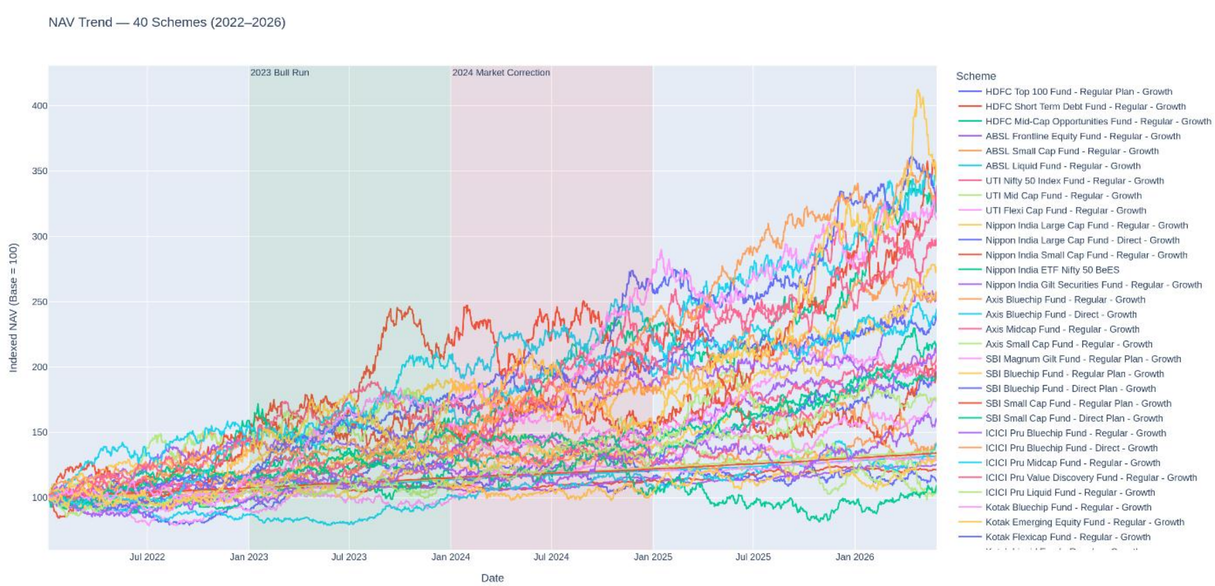
4. EDA Findings

The Exploratory Data Analysis phase was conducted to identify important patterns in **mutual fund NAV movements, AUM growth, SIP inflows, investor behaviour, folio growth, fund-return relationships, and portfolio sector allocation**. The analysis used Python libraries including Pandas, Matplotlib, Seaborn, and Plotly. The EDA covered the project's 40 mutual fund schemes and supporting industry and investor datasets.

4.1 NAV Trend Analysis

The daily NAV analysis across the selected mutual fund schemes showed clear differences in performance and volatility between fund categories. Equity-oriented schemes displayed larger fluctuations than relatively defensive schemes, reflecting their higher exposure to market movements.

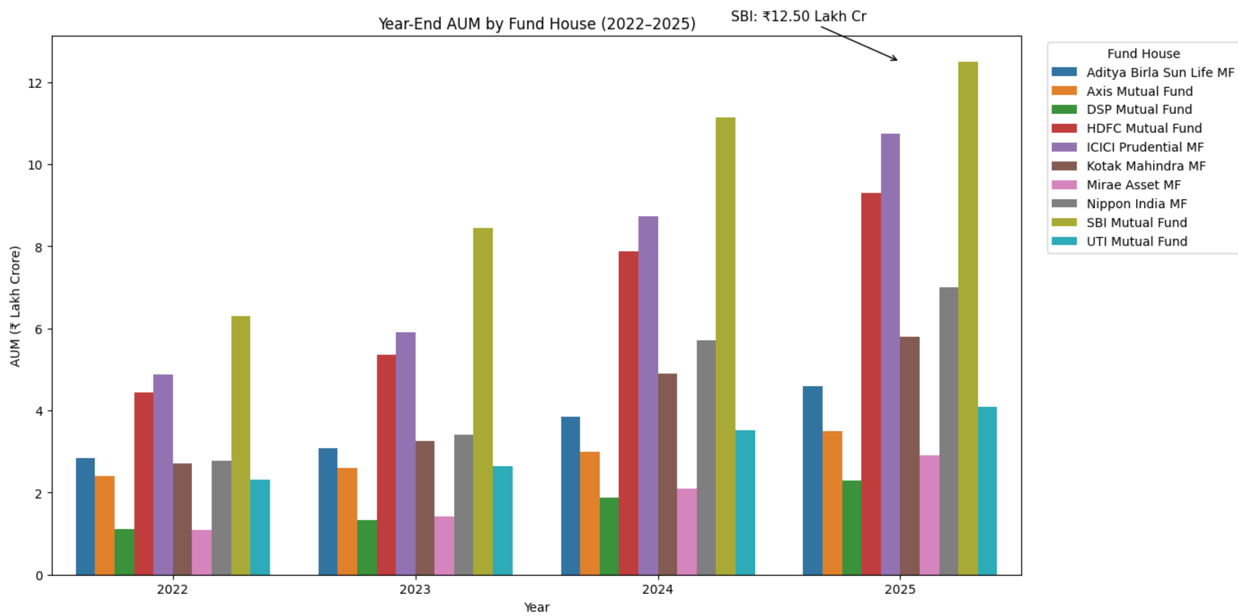
The NAV trend visualisation also highlights the major market phases during the analysis period, including the **2023 market rally and the corrections observed during 2024**.



4.2 AUM Growth by Fund House

The fund-house AUM analysis shows strong growth across the major AMCs during 2022–2025. **SBI Mutual Fund** recorded the highest AUM in the analysed fund-house dataset, reaching approximately **₹12.50 lakh crore**, demonstrating its dominant position among the selected fund houses.

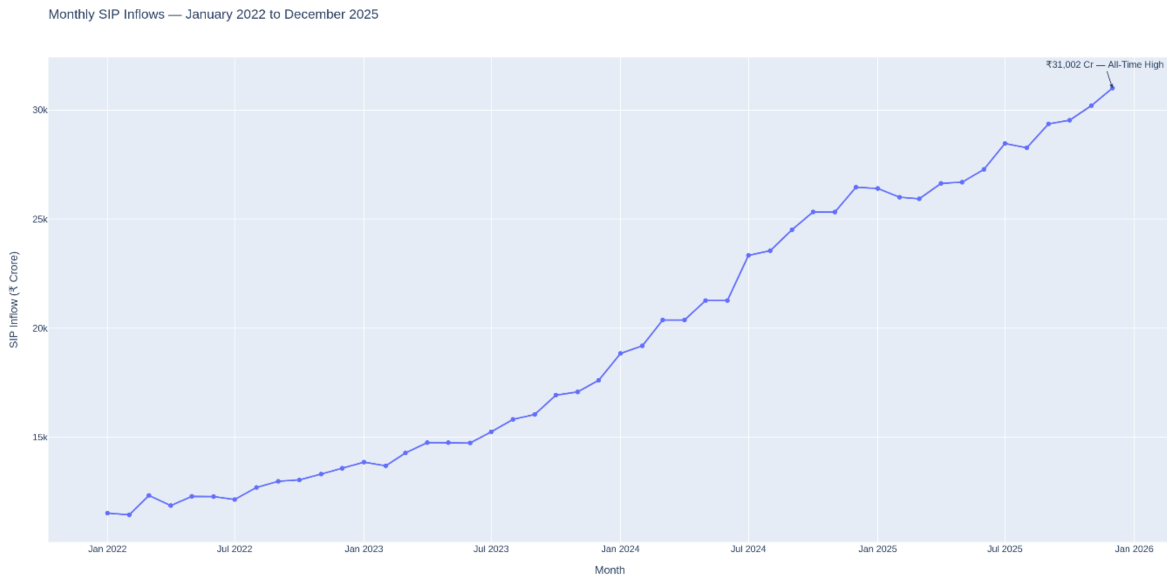
The analysis compares AUM across fund houses and years to identify changes in relative market size.



4.3 SIP Inflow Trend

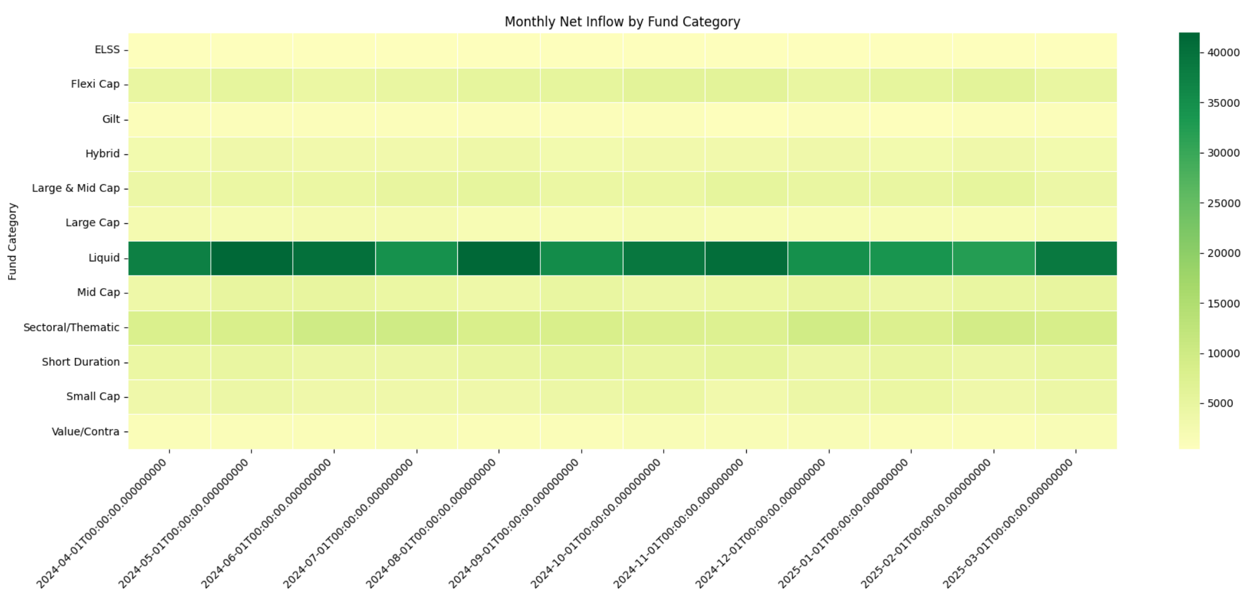
Monthly SIP inflows demonstrated a strong upward trend between January 2022 and December 2025. The series reached an **all-time high of ₹31,002 crore in December 2025**.

This growth indicates increasing adoption of systematic investing and demonstrates the growing importance of SIPs within the mutual fund ecosystem.



4.4 Category-wise Inflows

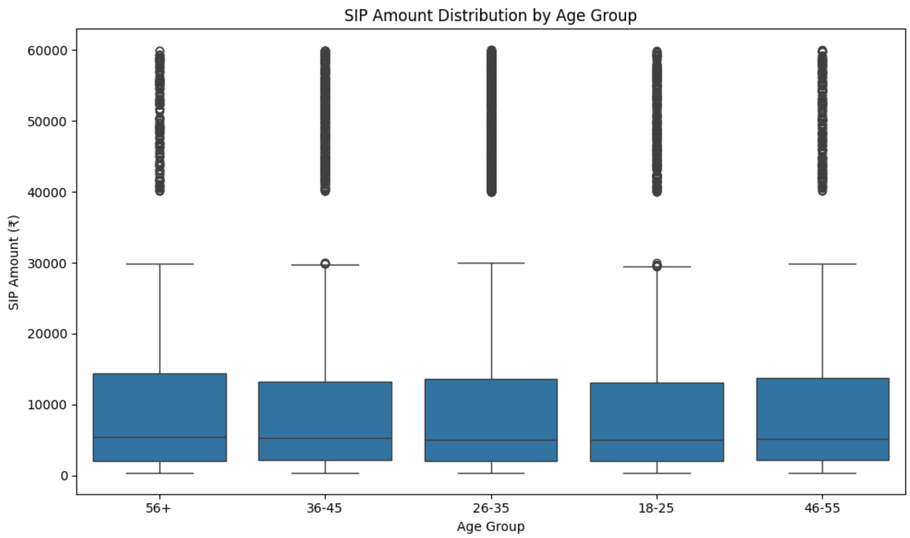
The category inflow heatmap was used to compare monthly net flows across mutual fund categories. The colour intensity makes it possible to identify periods of stronger inflows as well as categories experiencing comparatively weaker or negative flows.

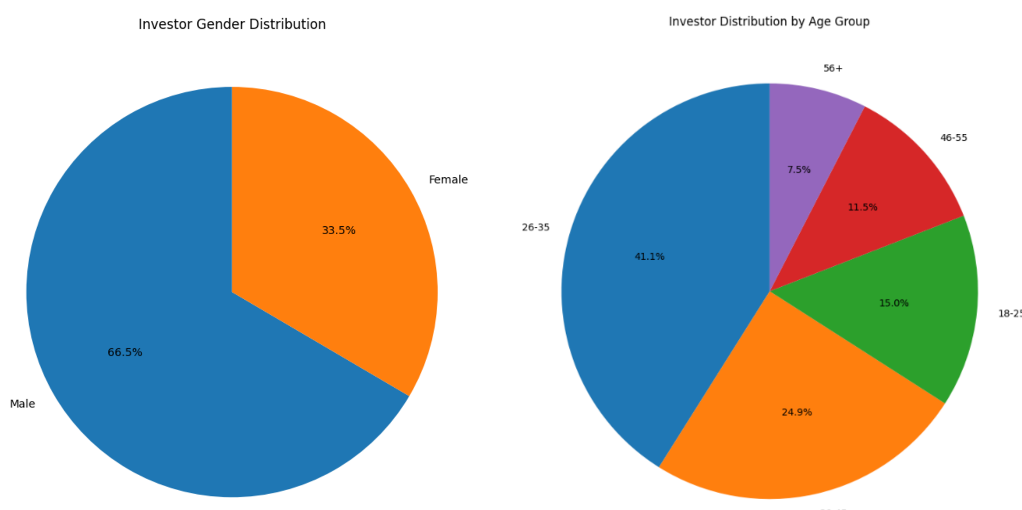


4.5 Investor Demographics

The investor dataset was analysed across age groups and gender to understand the demographic composition of the investor base.

The age distribution chart shows the relative contribution of each age group, while the SIP amount box plot highlights differences in the distribution of SIP investment amounts between age groups.

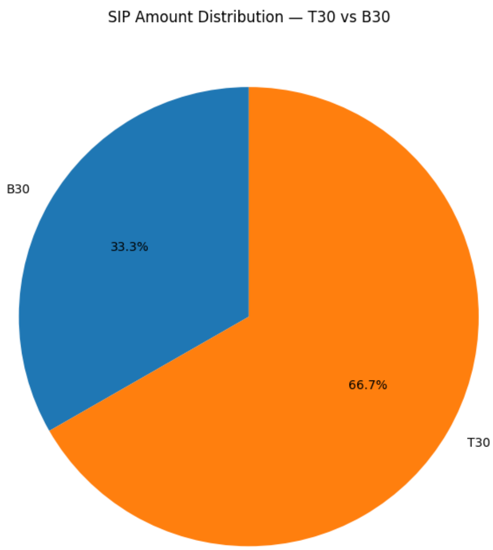


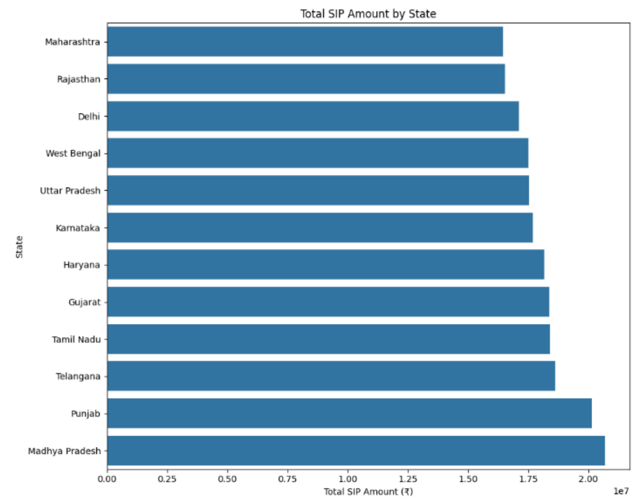


4.6 Geographic Distribution

The geographic analysis examined SIP amounts across Indian states and compared investment behaviour between **T30 and B30 cities**.

The state-wise horizontal bar chart identifies locations contributing higher SIP amounts, while the T30/B30 chart provides a high-level view of investment distribution between major cities and locations beyond the top 30 cities.

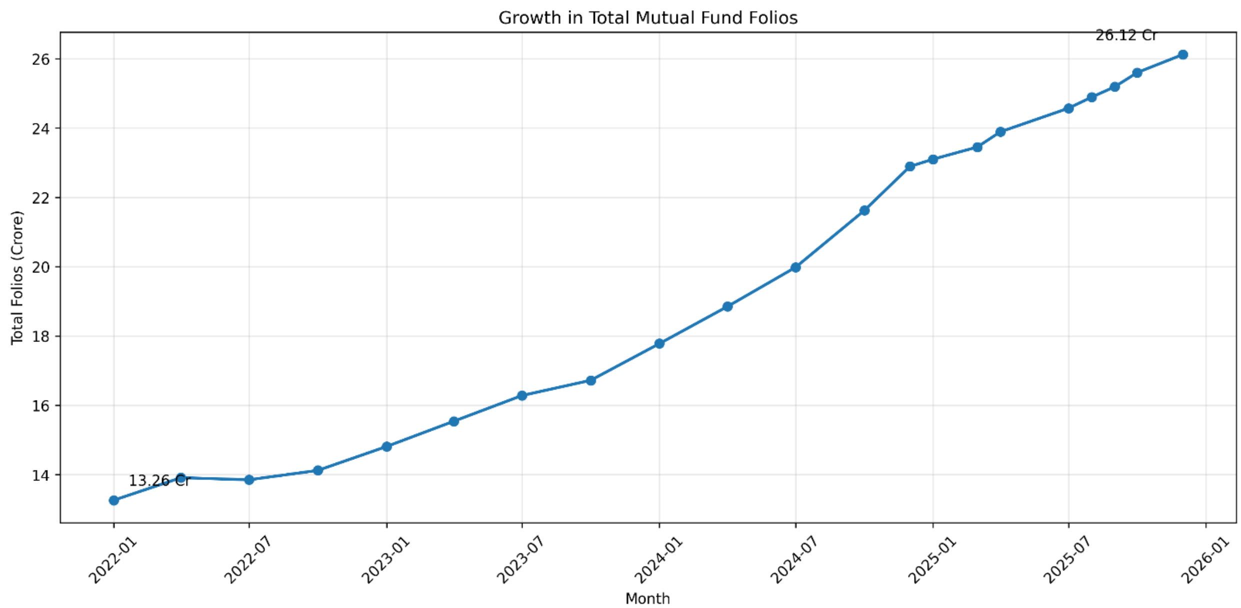




4.7 Folio Count Growth

The folio analysis shows a substantial increase in total mutual fund folios from **13.26 crore in January 2022 to 26.12 crore in December 2025**.

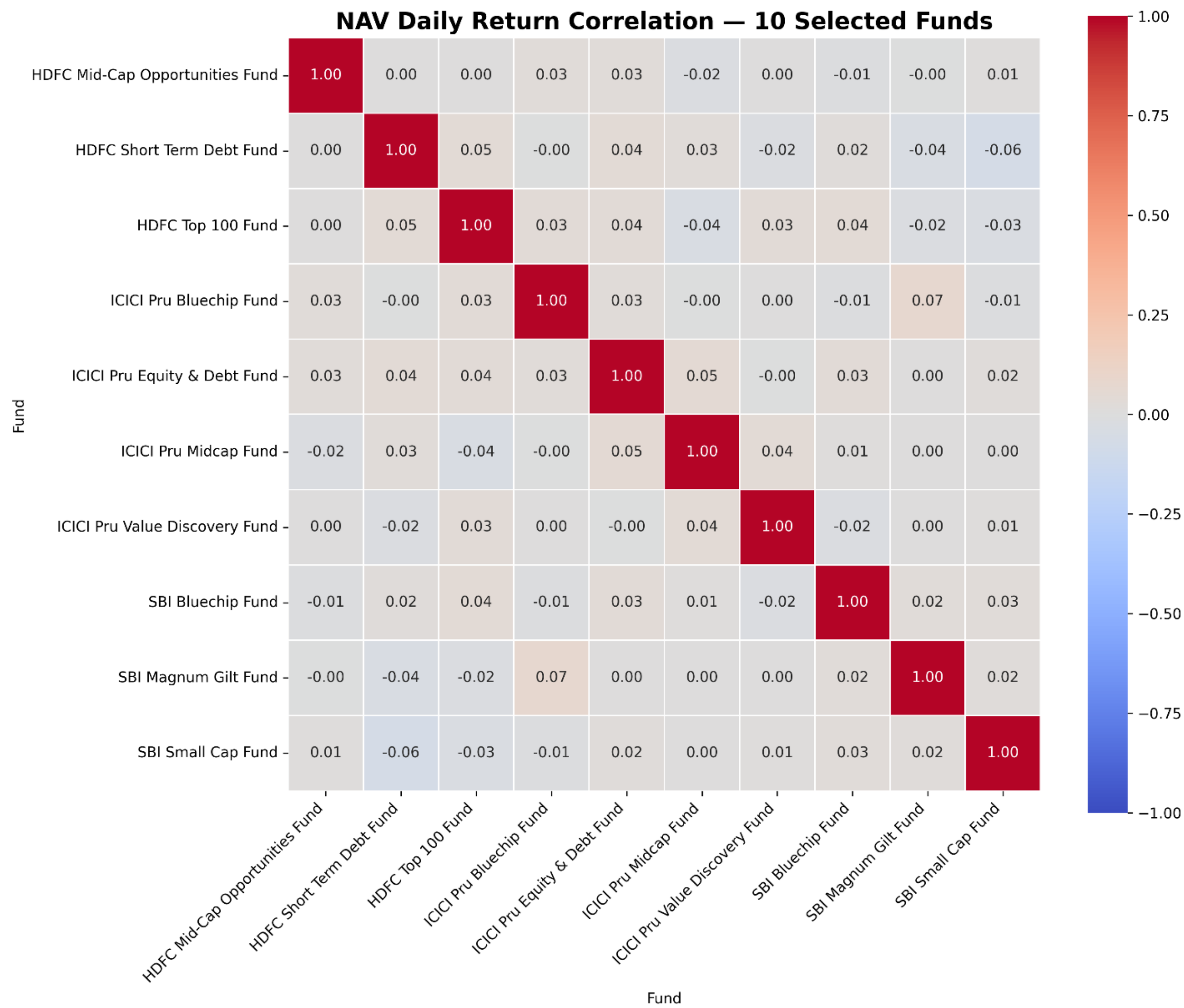
This represents an increase of approximately **96.98%**, indicating significant expansion in the mutual fund investor account base.



4.8 NAV Return Correlation

The correlation matrix compares daily returns across 10 selected mutual fund schemes.

The matrix helps identify funds whose returns move together and those that show relatively weaker relationships. Lower correlations can be useful when evaluating potential diversification opportunities.



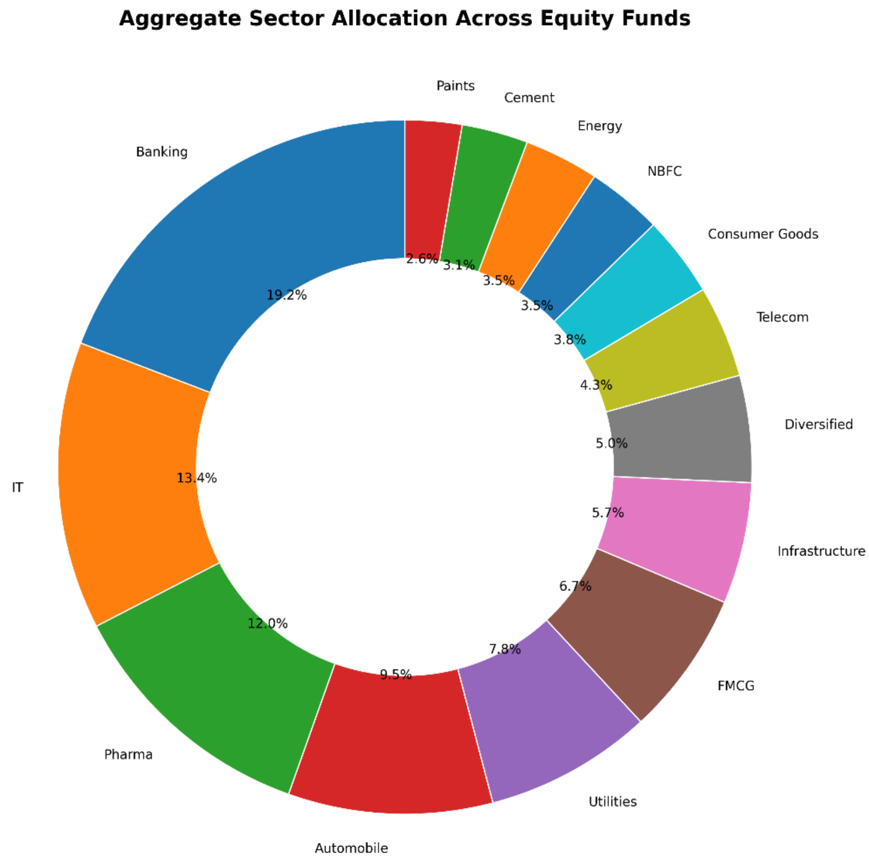
4.9 Sector Allocation

The sector allocation analysis aggregates portfolio holdings across the equity funds to identify the sectors receiving the highest portfolio weights.

The analysis identified:

- **Banking -19.18%**
- **IT -13.40%**

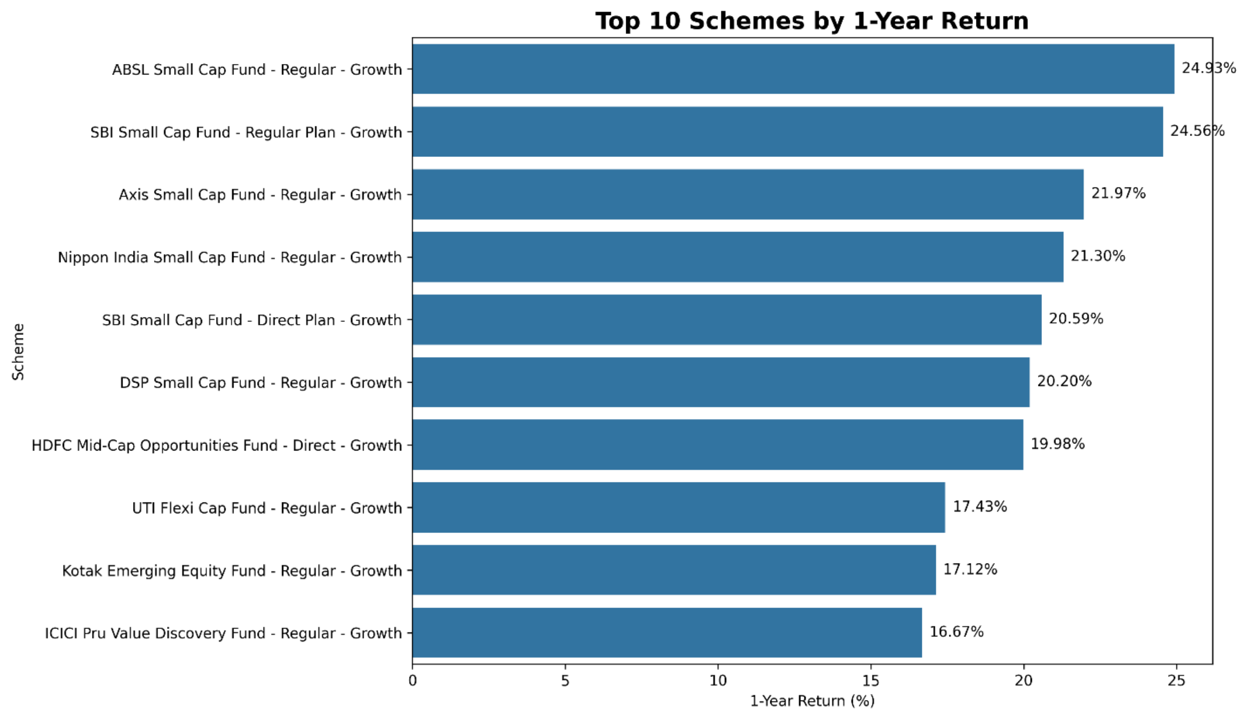
as the two largest sector allocations in the analyzed portfolio holdings.



4.10 Scheme Return Analysis

The 1-year return analysis highlighted several strong-performing schemes. The top performers identified in the analysis were:

Rank	Scheme	1-Year Return
1	ABSL Small Cap Fund -Regular -Growth	**24.93%**
2	SBI Small Cap Fund -Regular Plan -Growth	**24.56%**
3	Axis Small Cap Fund -Regular -Growth	**21.97%**
4	Nippon India Small Cap Fund -Regular -Growth	**21.30%**
5	SBI Small Cap Fund -Direct Plan -Growth	**20.59%**



4.11 Overall Conclusion

The EDA indicates a rapidly expanding mutual fund ecosystem characterised by strong SIP and folio growth, increasing AUM, differences in scheme-level risk and returns, varying investor behaviour, and meaningful sector and fund-house concentration. These findings establish the foundation for the subsequent performance and advanced analytics used to evaluate funds using risk-adjusted measures such as Sharpe, Sortino, Alpha, Beta, Maximum Drawdown, VaR, and CVaR.

5. Performance Analysis

The Performance Analysis phase evaluates the **40 mutual fund schemes** using return, risk-adjusted performance, market sensitivity, drawdown, and benchmark-comparison metrics. Daily NAV data containing **64,320 observations** was used to calculate daily returns and subsequent performance measures.

5.1 Daily Return Analysis

Daily returns were calculated using:

Daily Return = (NAV_t / NAV_{t-1}) − 1

The analysis produced **64,280 valid daily-return observations**. The average daily return was approximately **0.045%**, with a standard deviation of **0.87%**. The observed minimum and maximum daily returns were approximately **−5.81% and +6.47%**, respectively.

Finding: The daily-return distribution shows relatively small typical movements with occasional larger positive and negative observations, reflecting the market-driven nature of mutual fund NAVs.

5.2 CAGR Analysis

CAGR was calculated for **1-year, 3-year, and 5-year periods** across all 40 schemes.

Metric	Mean	Minimum	Maximum
1-Year CAGR	19.44%	−42.82%	82.85%
3-Year CAGR	16.41%	−11.70%	35.10%
5-Year CAGR	16.74%	1.17%	32.83%

All 40 schemes had sufficient observations for the three CAGR calculations.

Finding: Long-term returns varied substantially across schemes, with the strongest-performing funds generating significantly higher CAGR while a smaller number of schemes experienced negative shorter-term growth.

5.3 Sharpe Ratio

The Sharpe Ratio was calculated using a **6.5% annual risk-free rate**, converted to a daily rate and annualised using 252 trading days.

The highest Sharpe Ratio was recorded by **Mirae Asset Large Cap Fund -Regular - Growth**, with a Sharpe Ratio of approximately **1.07**.

Other highly ranked schemes included:

- Kotak Flexicap Fund -**0.97**
- Mirae Asset Tax Saver Fund -**0.92**
- ICICI Pru Midcap Fund -**0.88**
- SBI Bluechip Fund -**0.86**

Liquid and debt funds produced strongly negative Sharpe values because their returns were below the selected 6.5% risk-free benchmark.

Finding: Actively managed equity-oriented schemes dominated the highest Sharpe rankings, while low-volatility liquid and debt funds produced negative values under the selected risk-free-rate assumption.

5.4 Sortino Ratio

The Sortino Ratio evaluates returns against **downside volatility only**.

Mirae Asset Large Cap Fund again ranked first, with a Sortino Ratio of approximately **1.49**, followed by Kotak Flexicap Fund at approximately **1.48**.

The ranking was broadly consistent with the Sharpe Ratio results.

Finding: The consistency between Sharpe and Sortino rankings indicates that several of the leading equity funds delivered stronger returns relative to their overall and downside volatility.

The analysis also showed extremely negative Sortino values for some liquid funds. These values should be interpreted cautiously because their low volatility and limited number of negative-return observations can make downside deviation unstable.

5.5 Alpha and Beta Analysis

Alpha and Beta were calculated using **OLS regression against the Nifty 100**.

The analysis used approximately **1,149 benchmark trading-day observations**. However, the notebook identified an important data limitation: the fund NAV data contains forward-filled weekends/holidays, whereas the benchmark contains genuine trading days only.

For example, HDFC Top 100 showed:

- **Beta:** −0.058
- **Annualised Alpha:** 3.75%
- **R^2:** 0.0027

- **Fund/Nifty 100 return correlation:** −0.052

The very low R^2 and correlation indicate that the regression results should be interpreted cautiously.

Finding: The Alpha/Beta analysis was affected by the characteristics of the available benchmark and NAV data, resulting in unexpectedly low correlations and R^2 values. Therefore, these regression outputs are useful for demonstrating the analytical methodology but should not be interpreted as definitive measures of benchmark sensitivity.

5.6 Maximum Drawdown

Maximum Drawdown measures the largest decline from a previous NAV peak to a subsequent trough.

The largest drawdowns in the analysis included:

Scheme	Maximum Drawdown
SBI Small Cap Fund -Direct	**−52.57%**
Axis Small Cap Fund -Regular	**−51.68%**
ABSL Small Cap Fund -Regular	**−35.45%**
DSP Small Cap Fund -Regular	**−31.17%**
SBI Small Cap Fund -Regular	**−28.71%**

By comparison, liquid funds had very small drawdowns, with the lowest-risk schemes showing drawdowns of less than 0.2%.

Finding: Small-cap schemes experienced substantially larger drawdowns than liquid and debt funds, demonstrating the trade-off between higher return potential and greater downside risk.

5.7 Fund Scorecard

A composite **0–100 Fund Scorecard** was created using the following weights:

- **30%** -3-Year Return
- **25%** -Sharpe Ratio
- **20%** -Alpha
- **15%** -Expense Ratio

- **10%** -Maximum Drawdown

The top five funds were:

Rank	Fund	Score
1	ICICI Pru Midcap Fund -Regular -Growth	**80.63**
2	Kotak Flexicap Fund -Regular -Growth	**79.00**
3	SBI Small Cap Fund -Regular Plan -Growth	**76.25**
4	HDFC Mid-Cap Opportunities Fund -Regular -Growth	**76.25**
5	Mirae Asset Large Cap Fund -Regular -Growth	**75.00**

Finding: Mid-cap, flexi-cap, and small-cap schemes occupied most of the top positions, reflecting their stronger combination of long-term returns and risk-adjusted performance in the analysed dataset.

5.8 Benchmark Comparison

The top five scorecard funds were compared against the **Nifty 50 and Nifty 100** over a three-year period.

The selected top five were:

1. ICICI Pru Midcap Fund
2. Kotak Flexicap Fund
3. SBI Small Cap Fund
4. HDFC Mid-Cap Opportunities Fund
5. Mirae Asset Large Cap Fund

Tracking error versus Nifty 100 was calculated as the annualised standard deviation of the difference between fund and benchmark returns.

Fund	Tracking Error vs Nifty 100
ICICI Pru Midcap Fund	23.27%
Kotak Flexicap Fund	20.65%
SBI Small Cap Fund	28.67%

Fund	Tracking Error vs Nifty 100
HDFC Mid-Cap Opportunities Fund	22.50%
Mirae Asset Large Cap Fund	**18.80%**

Finding: Mirae Asset Large Cap Fund had the lowest tracking error among the five selected funds, while SBI Small Cap Fund showed the highest, reflecting greater divergence from the Nifty 100 benchmark.

5.9 Overall Conclusion

The performance analysis shows that higher-return mutual fund schemes generally came with greater downside risk and drawdown exposure. Risk-adjusted measures such as Sharpe, Sortino, Alpha, Beta, and Maximum Drawdown therefore provide a more comprehensive basis for evaluating schemes than return alone. The composite Fund Scorecard combines these dimensions to identify funds with a stronger overall performance profile.

6. Dashboard

The Power BI dashboard was developed as an interactive four-page analytical platform for exploring mutual fund industry trends, fund performance, investor behavior, and SIP/market trends. The dashboard includes interactive slicers and visualizations that allow users to filter and analyze the underlying data from multiple perspectives.

6.1 Industry Overview

The **Industry Overview** page provides a high-level summary of the mutual fund dataset.

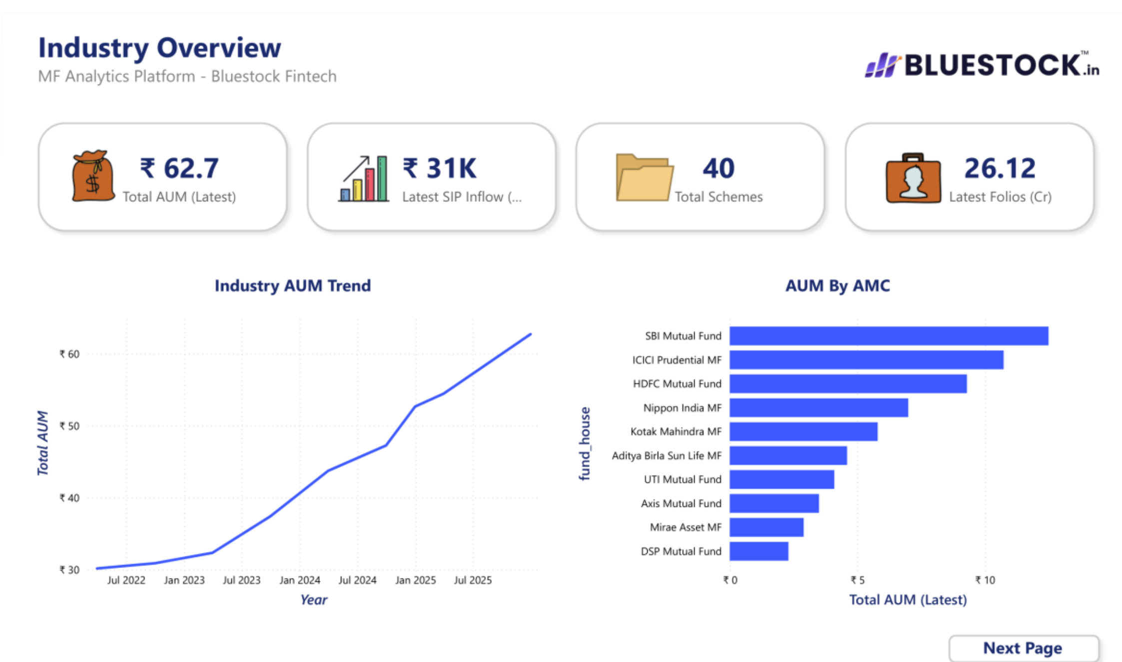
The page includes KPI cards for:

- Total AUM
- Latest SIP Inflow
- Total Schemes
- Latest Folio Count

It also contains an **Industry AUM Trend** line chart and an **AUM by AMC** bar chart.

The dashboard shows an upward trend in industry AUM over the analyzed period, while SBI Mutual Fund appears as the largest AMC by the latest AUM in the dashboard.

Figure 6.1: Industry Overview Dashboard



6.2 Fund Performance

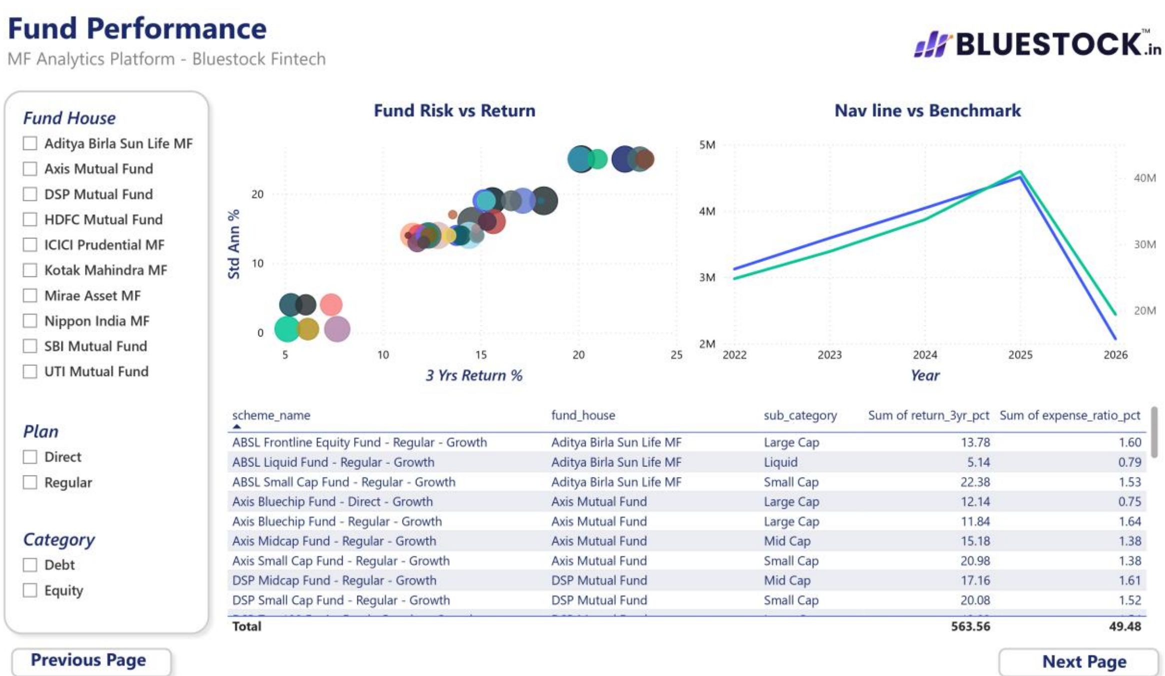
The **Fund Performance** page focuses on comparing mutual fund schemes based on their return and risk characteristics.

The main visualizations include:

- **Fund Risk vs Return** scatter plot
- **NAV Line vs Benchmark**
- Fund performance table
- Fund House slicer
- Plan slicer
- Category slicer

The risk-return scatter plot allows users to identify funds with relatively higher returns and their associated volatility. The performance table provides scheme-level details including 3-year return and expense ratio.

Figure 6.2: Fund Performance Dashboard



scheme_name	fund_house	sub_category	Sum of return_3yr_pct	Sum of expense_ratio_pct
ABSL Frontline Equity Fund - Regular - Growth	Aditya Birla Sun Life MF	Large Cap	13.78	1.60
ABSL Liquid Fund - Regular - Growth	Aditya Birla Sun Life MF	Liquid	5.14	0.79
ABSL Small Cap Fund - Regular - Growth	Aditya Birla Sun Life MF	Small Cap	22.38	1.53
Axis Bluechip Fund - Direct - Growth	Axis Mutual Fund	Large Cap	12.14	0.75
Axis Bluechip Fund - Regular - Growth	Axis Mutual Fund	Large Cap	11.84	1.64
Axis Midcap Fund - Regular - Growth	Axis Mutual Fund	Mid Cap	15.18	1.38
Axis Small Cap Fund - Regular - Growth	Axis Mutual Fund	Small Cap	20.98	1.38
DSP Midcap Fund - Regular - Growth	DSP Mutual Fund	Mid Cap	17.16	1.61
DSP Small Cap Fund - Regular - Growth	DSP Mutual Fund	Small Cap	20.08	1.52
Total			563.56	49.48

6.3 Investor Analytics

The **Investor Analytics** page examines investor transaction behavior across demographic and geographic dimensions.

The page contains:

- Transaction amount by state
- Age group versus average SIP amount
- Monthly transaction volume
- Transaction amount by transaction type
- State slicer
- Age group slicer
- City-tier slicer

The dashboard indicates that the **26–35 age group has the highest average SIP amount** among the displayed age groups. Transaction activity also varies considerably across states.

The transaction-type donut chart provides a breakdown of the total transaction amount across the available transaction categories.

Figure 6.3: Investor Analytics Dashboard

6.4 SIP & Market Trends

The **SIP & Market Trends** page combines SIP inflow trends with market movement information.

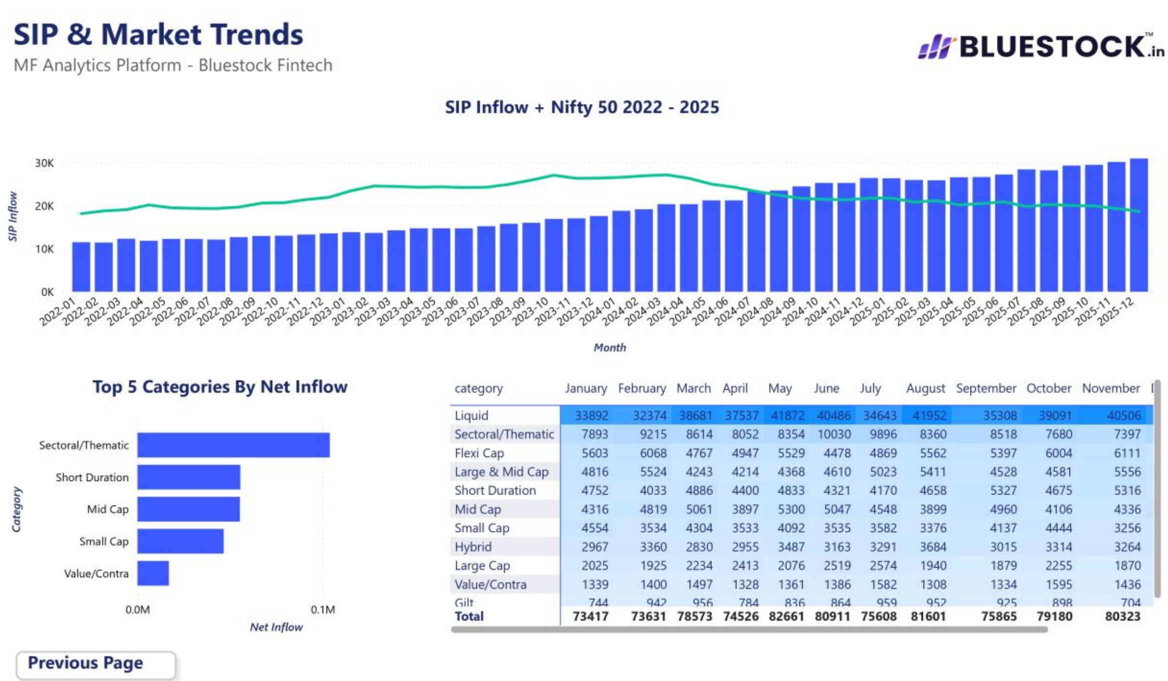
The main visualizations include:

- Monthly SIP inflow + Nifty 50 trend
- Top 5 categories by net inflow
- Category-level monthly inflow heatmap/table

The monthly SIP trend shows a strong increase over the 2022–2025 period, reaching approximately **₹31,002 crore in December 2025**, which represents the highest SIP inflow highlighted in the analysis.

The category analysis also shows differences in net inflows across categories, with **Sectoral/Thematic** appearing among the leading categories in the displayed ranking.

Figure 6.4: SIP & Market Trends Dashboard



category	January	February	March	April	May	June	July	August	September	October	November
Liquid	33892	32374	38681	37537	41872	40486	34643	41952	35308	39091	40506
Sectoral/Thematic	7893	9215	8614	8052	8354	10030	9896	8360	8518	7680	7397
Flexi Cap	5603	6068	4767	4947	5529	4478	4869	5562	5397	6004	6111
Large & Mid Cap	4816	5524	4243	4214	4368	4610	5023	5411	4528	4581	5556
Short Duration	4752	4033	4886	4400	4833	4321	4170	4658	5327	4675	5316
Mid Cap	4316	4819	5061	3897	5300	5047	4548	3899	4960	4106	4336
Small Cap	4554	3534	4304	3533	4092	3535	3582	3376	4137	4444	3256
Hybrid	2967	3360	2830	2955	3487	3163	3291	3684	3015	3314	3264
Large Cap	2025	1925	2234	2413	2076	2519	2574	1940	1879	2255	1870
Value/Contra	1339	1400	1497	1328	1361	1386	1582	1308	1334	1595	1436
Gilt	744	942	956	784	836	864	959	952	925	898	704
Total	**73417**	**73631**	**78573**	**74526**	**82661**	**80911**	**75608**	**81601**	**75865**	**79180**	**80323**

6.5 Dashboard Interactivity

The dashboard incorporates interactive filtering to allow users to explore the data dynamically.

The available filters include:

Dashboard Page	Interactive Filters
Industry Overview	Industry-level analysis
Fund Performance	Fund House, Plan, Category
Investor Analytics	State, Age Group, City Tier
SIP & Market Trends	Time/category analysis

The Fund Performance page enables users to narrow the analysis by **fund house, investment plan, and category**, while the Investor Analytics page provides filters for **state, age group, and city tier**.

This interactivity makes the dashboard suitable for exploratory analysis rather than presenting only static results.

7. Limitations

Although the Bluestock Mutual Fund Analytics Platform provides a comprehensive analytical view of mutual fund data, certain limitations should be considered when interpreting the results.

1. **Dataset coverage** -The analysis is based on the provided datasets and selected mutual fund schemes; therefore, the findings may not represent the complete Indian mutual fund industry.
2. **Historical analysis** -Most analysis is based on historical data from approximately 2022–2025/2026. Past performance does not guarantee future performance.
3. **Benchmark limitations** -Benchmark comparisons depend on the selected market indices and available benchmark data. Different benchmarks may produce different performance interpretations.
4. **Risk-free rate assumption** -The Sharpe Ratio uses **6.5% as a risk-free rate proxy**. Changes in this assumption can affect the resulting Sharpe rankings.
5. **Market data limitations** -NAV observations are subject to trading-day availability. Non-trading days such as weekends and holidays require appropriate handling during time-series analysis.
6. **Investor dataset limitations** -Investor demographic and transaction findings depend on the available transaction records and fields such as age group, gender, state and city tier.
7. **Risk metrics are historical** -Measures such as VaR, CVaR, maximum drawdown, volatility and Sortino Ratio describe historical risk and may not accurately predict future losses.
8. **Portfolio allocation limitations** -Sector allocation and HHI calculations are based on the holdings available in the portfolio dataset and therefore represent the dataset's observation period.
9. **Dashboard scope** -The Power BI dashboard focuses on the four defined analytical areas: industry overview, fund performance, investor analytics, and SIP/market trends.
10. **No investment advice** -The analytical results are intended for educational and analytical purposes and should not be treated as personalized investment recommendations.

8. Recommendations

Based on the EDA, fund performance, investor behaviour, risk analysis, and dashboard findings, the following recommendations are proposed for Bluestock Fintech.

8.1 Prioritise SIP Retention

The investor analysis identifies SIP continuity as an important business opportunity. Investors with six or more SIP transactions were analysed based on the gap between successive SIP transactions, with investors having an average gap above 35 days classified as **"at-risk."**

Recommendation:
Bluestock should develop an **SIP retention monitoring system** that identifies investors with increasing SIP gaps and enables targeted reminders, engagement campaigns, and retention interventions.

Business impact: Better SIP continuity can help maintain recurring investment flows and improve investor retention.

8.2 Build Risk-Based Fund Discovery

The analysis shows substantial differences in fund returns, volatility, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. The current fund scorecard already combines several of these metrics into a **0–100 composite score**.

Recommendation:
Bluestock should make **risk-adjusted fund discovery** a core feature of the platform rather than ranking schemes only by historical return.

Users should be able to filter funds by:

- Risk grade
- Return
- Sharpe Ratio
- Sortino Ratio
- Maximum Drawdown
- Expense Ratio
- Alpha

Business impact: This directly addresses the performance-comparison gap and makes fund selection more analytical and transparent.

8.3 Strengthen Benchmark-Based Fund Monitoring

The project compares fund returns against **Nifty 50, Nifty 100, and other relevant indices**, including tracking error and Alpha/Beta analysis.

Recommendation:
Every fund detail page should show:

Fund Return → Benchmark Return → Outperformance → Alpha → Beta → Tracking Error

This would allow users to understand whether a fund's performance is genuinely competitive relative to its benchmark.

Business impact: This directly addresses the project's **benchmark tracking problem (P3)**.

8.4 Target Investor Segments Using Behavioural Analytics

The investor dataset contains **age group, gender, state, city tier, transaction type, payment mode and KYC status**, providing a strong foundation for customer segmentation.

Your EDA also analyses:

- Age-group distribution
- Average SIP amount by age
- Gender distribution
- SIP amount by state
- T30 vs B30 participation

Recommendation:
Bluestock should create investor segments based on **investment amount, age, geography, transaction behaviour, and SIP activity**, then use these segments for targeted communication and product discovery.

Business impact: This can improve customer acquisition and engagement while addressing the project's **Investor Behaviour Blind Spot (P4)**.

8.5 Focus Acquisition on High-Value Investor Segments

The demographic analysis indicates differences in SIP contribution between age groups, while the geographic analysis shows significant variation in investment amounts across states and city tiers.

Recommendation:
Bluestock should identify segments with higher average SIP contributions and prioritize them for:

- SIP campaigns
- Financial education
- Goal-based investment content
- Digital acquisition
- Personalized fund discovery

Rather than using a single nationwide strategy, campaigns should be **segment-specific**.

8.6 Use SIP and Category Flows as Market Intelligence

The analysis tracks monthly SIP inflows and category-level net flows. SIP inflows reached **₹31,002 crore in December 2025**, while category analysis identifies differences in investor preference across fund categories.

Recommendation:
Bluestock should build a **market-trend monitoring layer** that flags:

- Record SIP inflows
- Rapid category inflow changes
- Declining categories
- Changes in investor preferences
- Growth in SIP accounts

Business impact: This can help management and distributors identify emerging market trends earlier.

8.7 Introduce Portfolio Concentration Alerts

The sector analysis aggregates equity-fund holdings, while the advanced analytics component calculates **Herfindahl-Hirschman Index (HHI)** to measure sector concentration.

Your analysis identified **Banking at 19.18%** and **IT at 13.40%** as the two largest aggregate sector allocations.

Recommendation:
Bluestock should display a **sector concentration indicator** alongside fund recommendations and portfolio analysis.

For highly concentrated portfolios, the dashboard could show:

Low concentration → Moderate concentration → High concentration

Business impact: This provides users with an additional perspective on diversification and portfolio risk.

8.8 Monitor AUM and AMC Market Position

The EDA tracks AUM for the 10 largest fund houses from 2022–2025 and highlights **SBI Mutual Fund's approximately ₹12.50 lakh crore AUM** in December 2025.

Recommendation:
Bluestock should monitor:

- AMC AUM growth
- Market-share movement
- Year-over-year AUM change
- Number of schemes
- Relative AMC performance

Business impact: This can provide management with a competitive view of the mutual fund industry.

8.9 Make the Dashboard the Primary Business Reporting Layer

The current dashboard already provides four analytical perspectives:

Industry Overview → Fund Performance → Investor Analytics → SIP & Market Trends

It also includes slicers, tooltips and drill-through functionality.

Recommendation:
Bluestock should use the Power BI dashboard as the **central self-service reporting layer** instead of relying primarily on static monthly reports.

The dashboard should be connected to the refreshed ETL output so that updated data automatically flows into the analytical views.

Business impact: This directly addresses **P5 -Slow Reporting**.

9. Conclusion

The **Bluestock Mutual Fund Analytics Platform** successfully demonstrates an end-to-end data analytics solution for the Indian mutual fund domain. The project integrates data ingestion, cleaning, transformation, storage, exploratory analysis, performance measurement, advanced risk analytics, and interactive visualization into a single analytical workflow.

The ETL pipeline consolidates multiple datasets covering **NAV history, AUM, SIP inflows, folio counts, fund performance, investor transactions, portfolio holdings, and benchmark indices**. This addresses the data-fragmentation problem by creating a structured and reusable analytical data layer.

The EDA identified important industry and investor trends, including continued growth in mutual fund folios, increasing SIP participation, differences in AMC-level AUM, category-level flow patterns, investor demographic variations, and sector concentration. The analysis also produced more than 15 visualizations to support these findings.

The performance analytics component evaluated the 40 selected schemes using **CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, volatility, maximum drawdown, and benchmark comparison**. A composite fund scorecard was also developed to provide a structured method for comparing schemes across multiple performance and risk dimensions.

The advanced analytics extended the project beyond traditional performance reporting by incorporating **VaR, CVaR, rolling Sharpe Ratio, investor cohort analysis, SIP continuity analysis, fund recommendation logic, and sector concentration using HHI**.

The Power BI dashboard brings these analytical outputs together through four major views: **Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends**. Interactive filters, drill-through functionality, KPI cards, and comparative visualizations provide users with a self-service environment for exploring mutual fund data.